Jeevan Kumar Banoth - 02105145

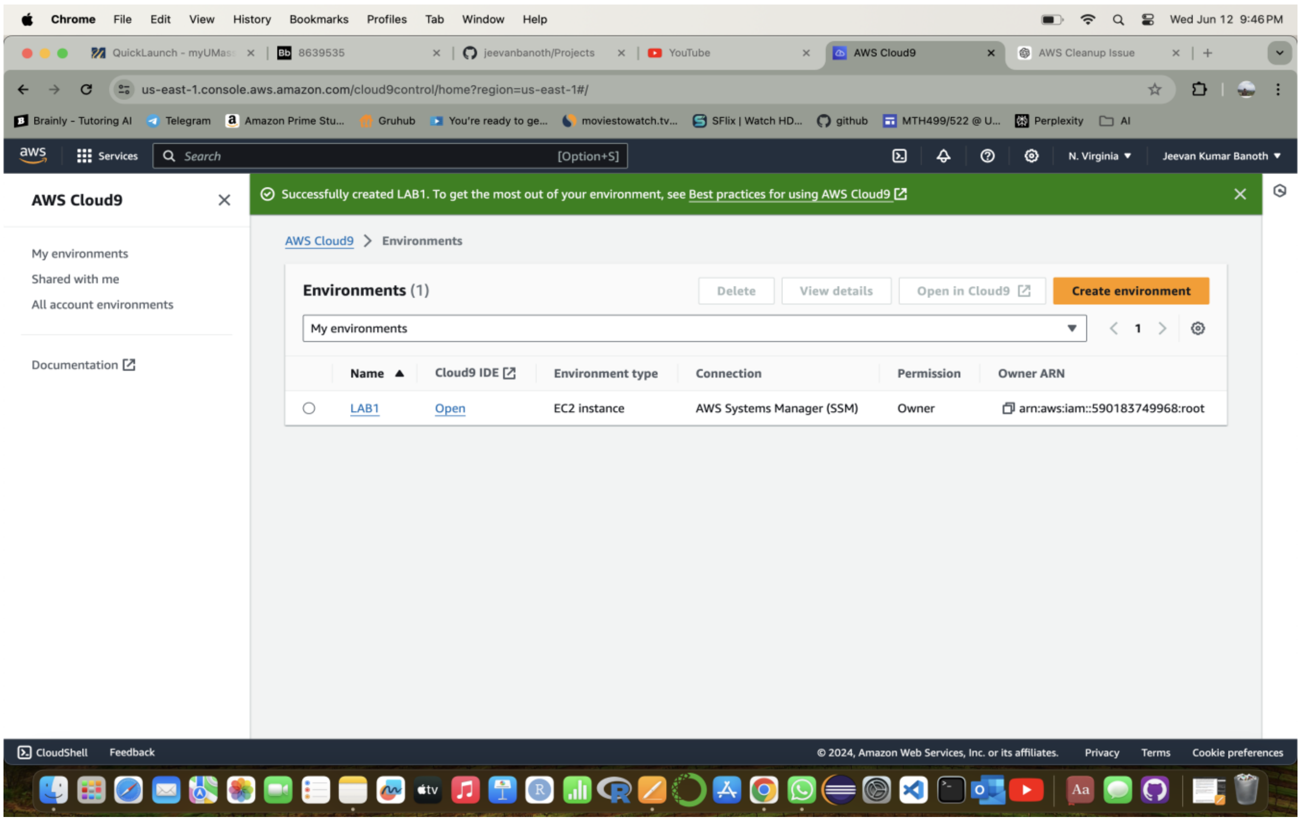
CIS 602-7103: Special Topics in CIS

Homework – 1:

Accessing and Analyzing Data by Using Amazon S3: -

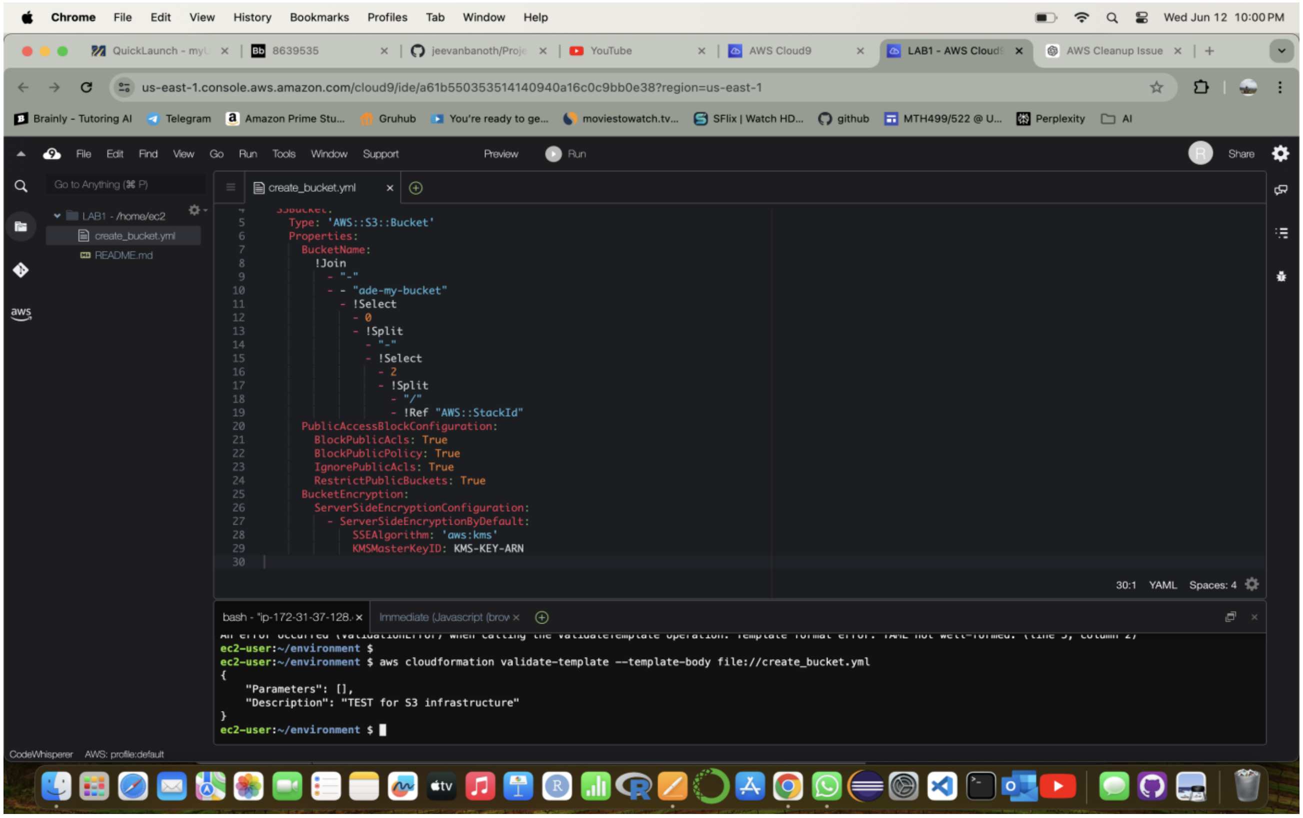
Task 1: Creating a CloudFormation template and stack

→ I tried accessing the AWS academy student version but due to some region issues the Lab is not starting in there so I will be doing my Lab in the AWS academy that I created on my own.



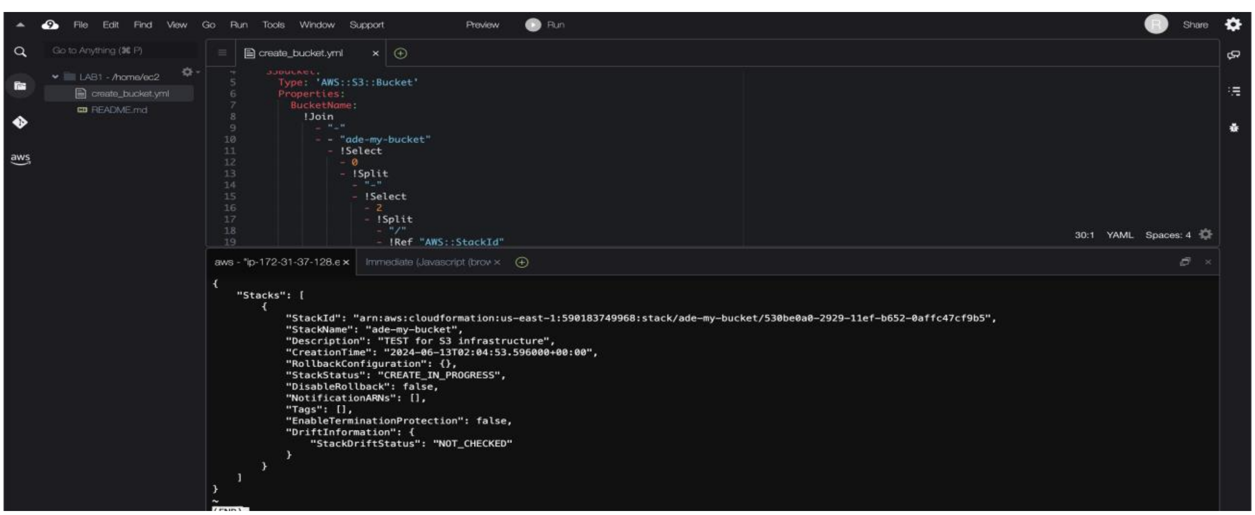
Step 1: Set Up AWS Cloud9 Environment: -

By using the Cloud9 formation in the AWS, we will be creating an .yml template file and saving the file as create_bucket.yml and running the commands which are given.

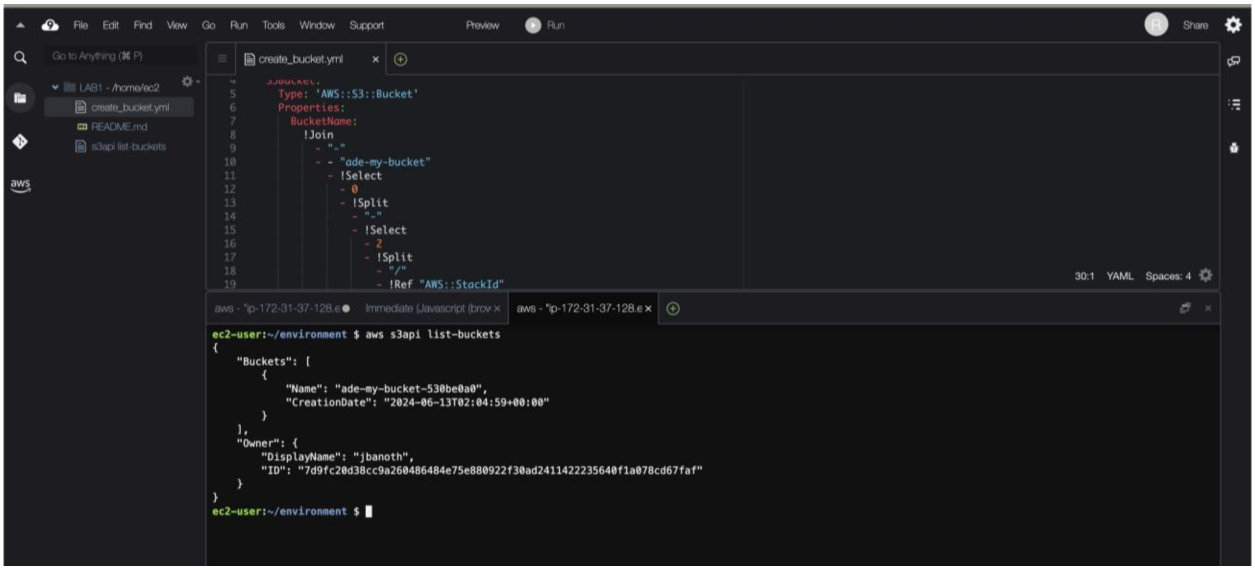


By using the commands given in the manual file, I have validated the in the AWS cloud9 formation template as shown above and it shows that template has been validated correctly.

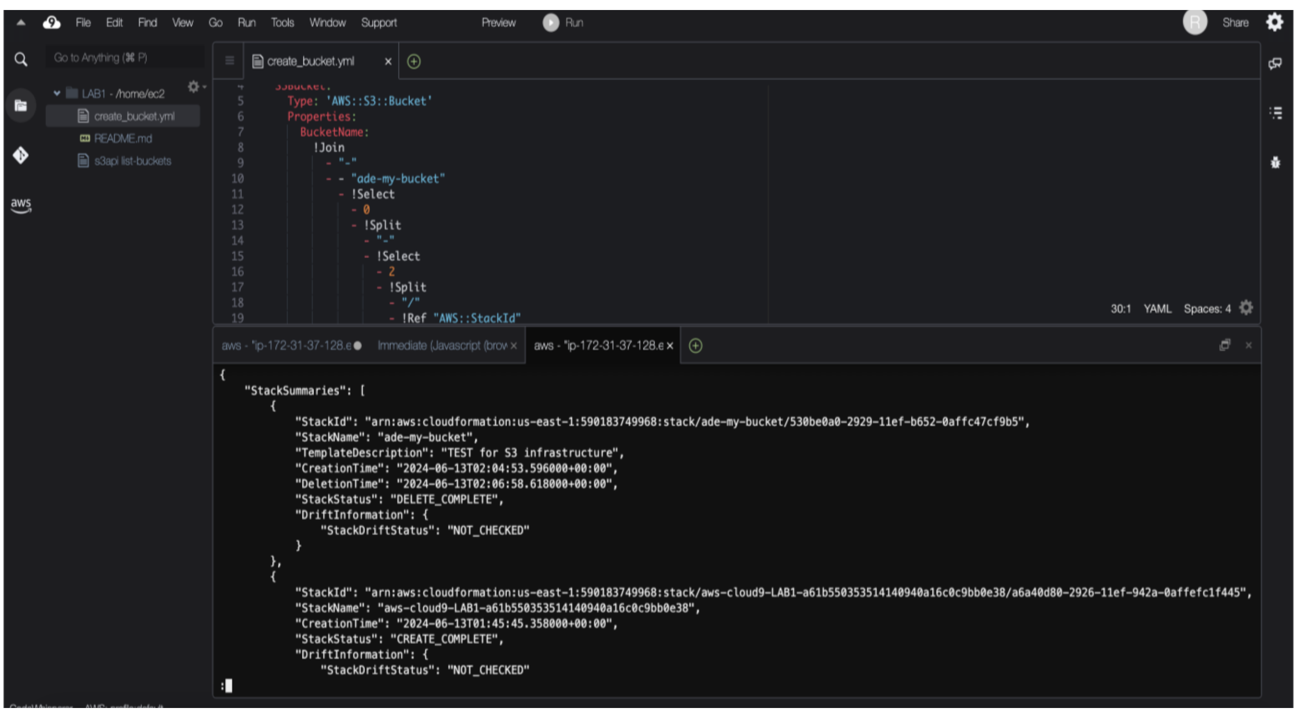
Next, we will be creating the stack using the command given in the manual and replacing it with the stack name below.



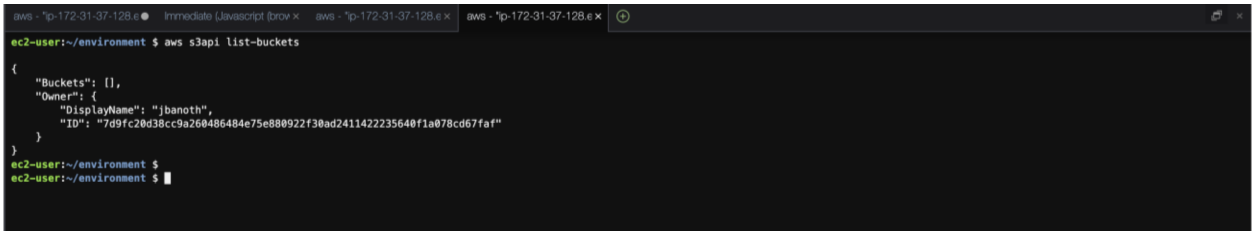
As we can see over there, stack has been created successfully. I have also used the commands which verify that stack has been created or not which are shown in the below image.



Now, by using the new command tab, I have deleted the stack using the command prompt which is given below, and verified whether that stack has been deleted successfully or not and that it came out as below one.



As the cursor has been stuck over there, I will be using the new terminal to check how many buckets are there.



As we have deleted all my stack which I have created just now, the list in the bucket shows empty.

SUMMARY OF THE TASK 1:

So, in this task we have created cloud9 environment template as given and developed a template for an S3 bucket.

We have validated the template using AWS CLI.

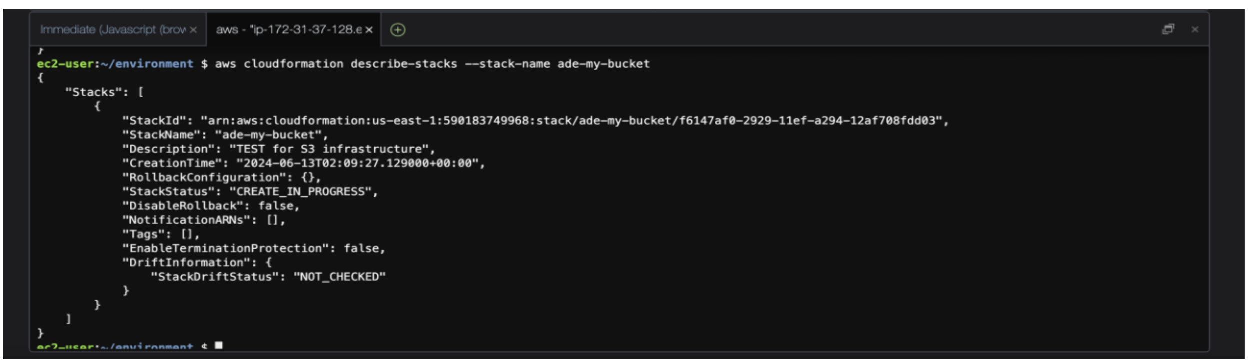
Deployed the stack using terminal window in cloud9 and verified the S3 bucket.

Also, by using the same terminal we have deleted the stack which is created and verified if the stack has been successfully deleted or not.

Next, we checked how many buckets are present in the environment and summarized it.

Task 2: Uploading sample data to an S3 bucket: -

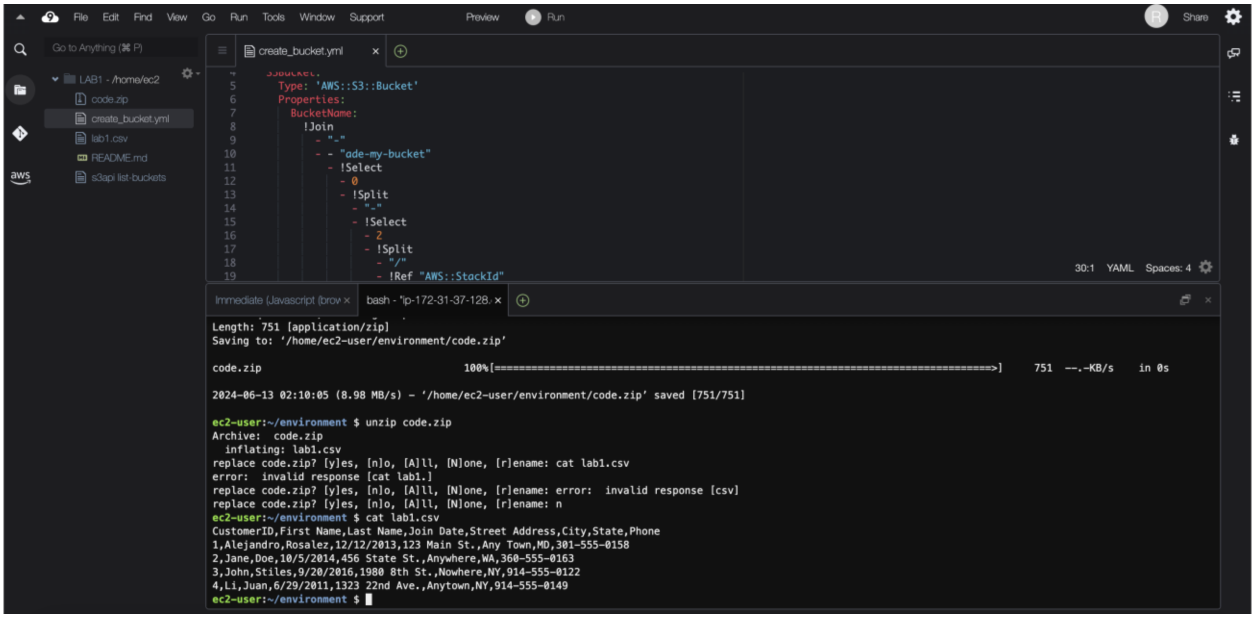
As we have deleted all the stacks which have been created, we will be recreating it by using the same command which is used before.



In cloud9 terminal, we have used the given commands for downloading and unzipping the given file as below.

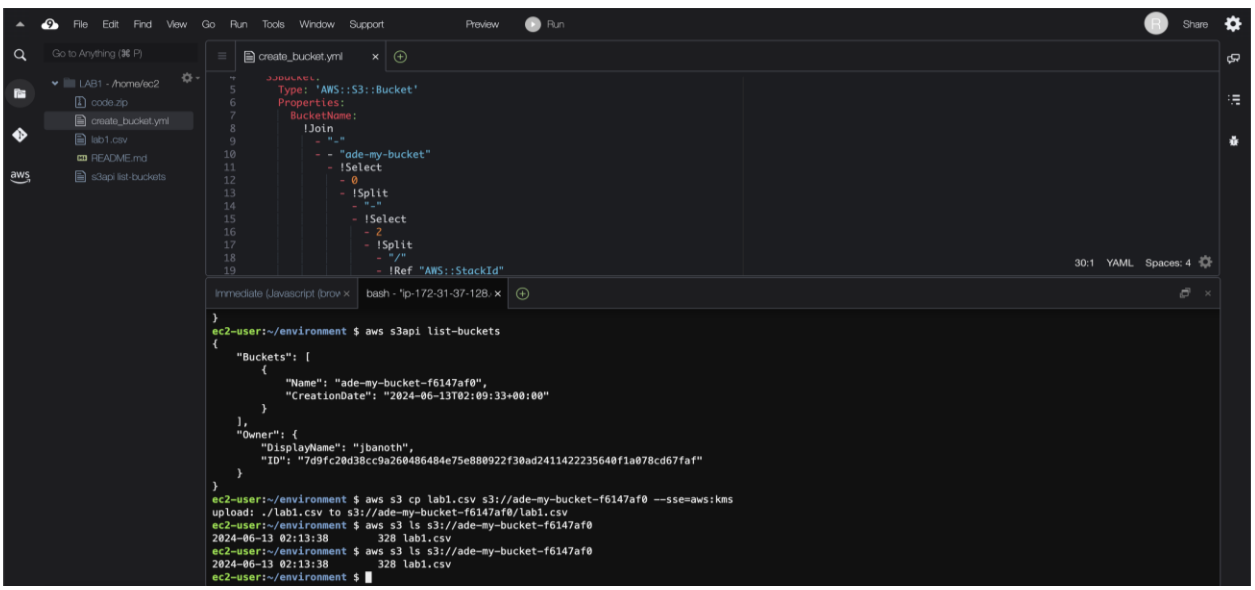
When prompted to unzip the code.zip file we need to enter n and return it.

To view the contents which are present in the file, we will be using the cat command and as shown below.



Next, we will be listing all the buckets which are present. And copying the dataset to S3 bucket.

Then verifying the upload is successful or not.



Task 2 summary: -

In this task, we have recreated S3 bucket using CloudFormation template.

Downloaded and unzipped the sample data file which is there. And examined the data in the csv file which is unzipped.

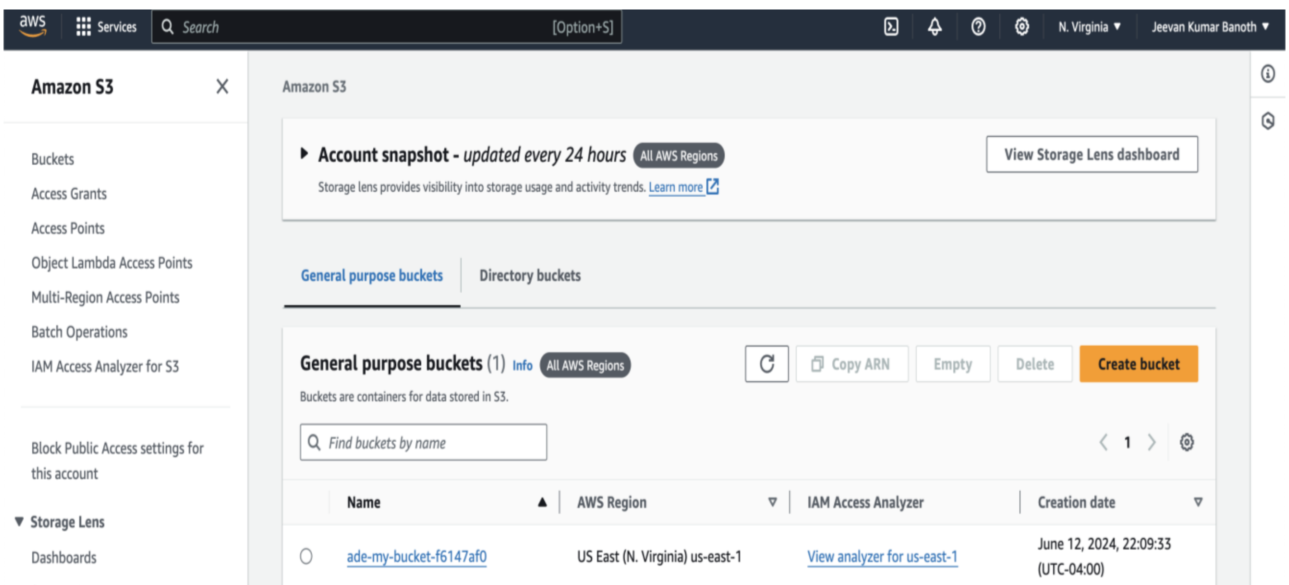
Then I used AWS CLI commands to copy the dataset file which is downloaded into the S3 bucket.

Next, we verified if the copying dataset into the S3 bucket was done successfully or not.

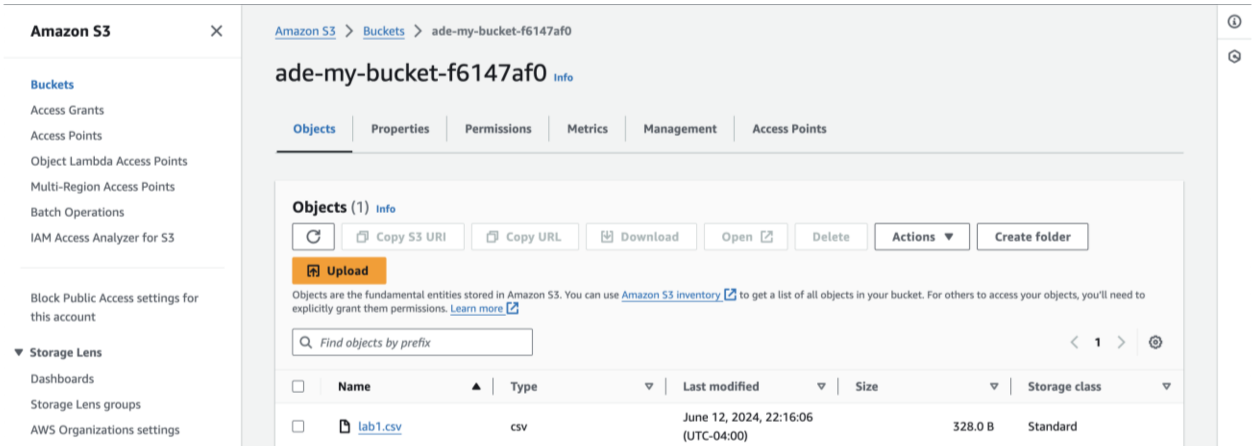
This exercise will clarify how to use Amazon S3 bucket and perform small operations like creating the stack and managing the dataset.

Task 3: Querying the data with S3 select: -

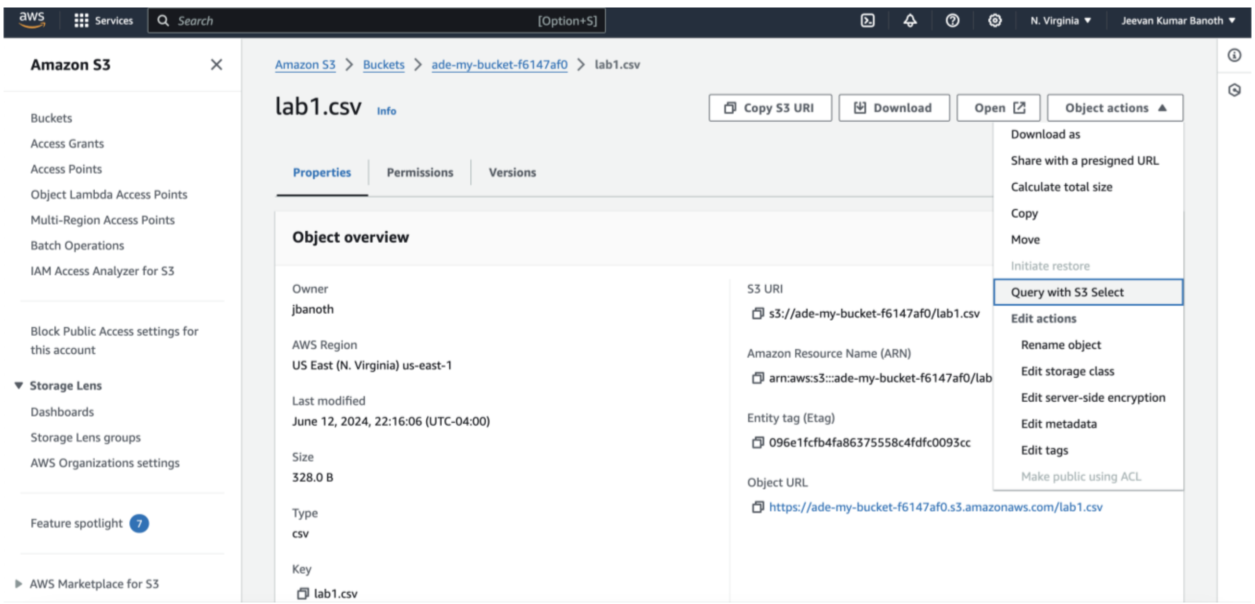
Now, we will be accessing the S3 bucket in AWS management console for that first, we open the AWS management console and open the S3 bucket from the search bar, in the bucket list, we will be opening the existing bucket.



Inside the bucket, I will be selecting the data file which is lab1.csv, next in the Object actions choosing Query with S3 select.



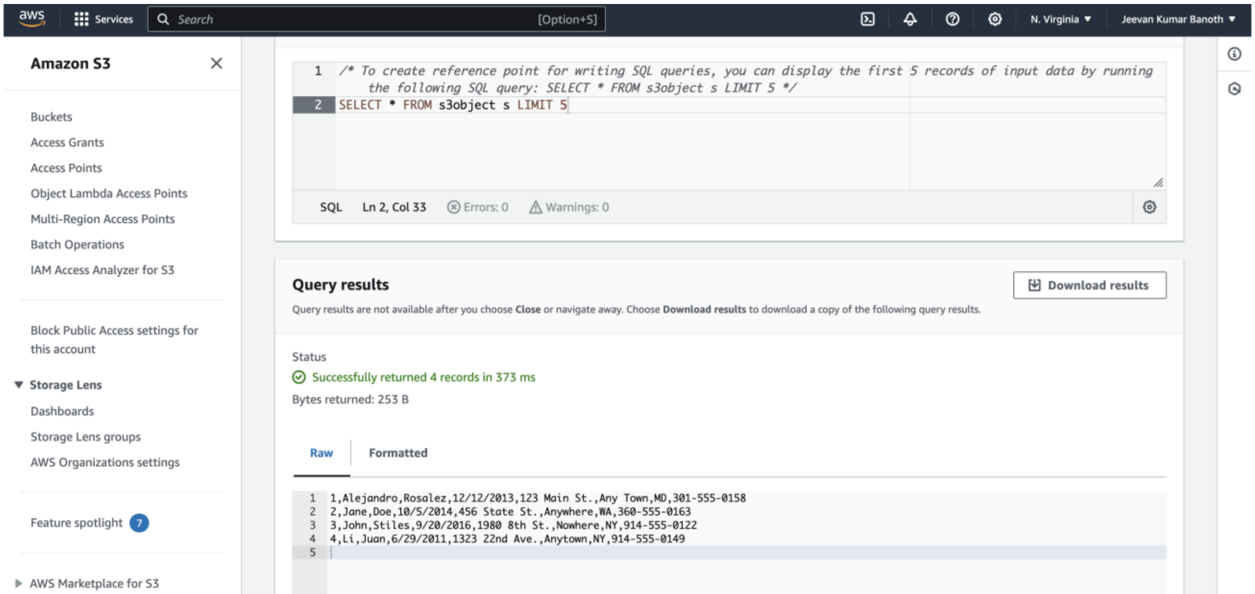
↓



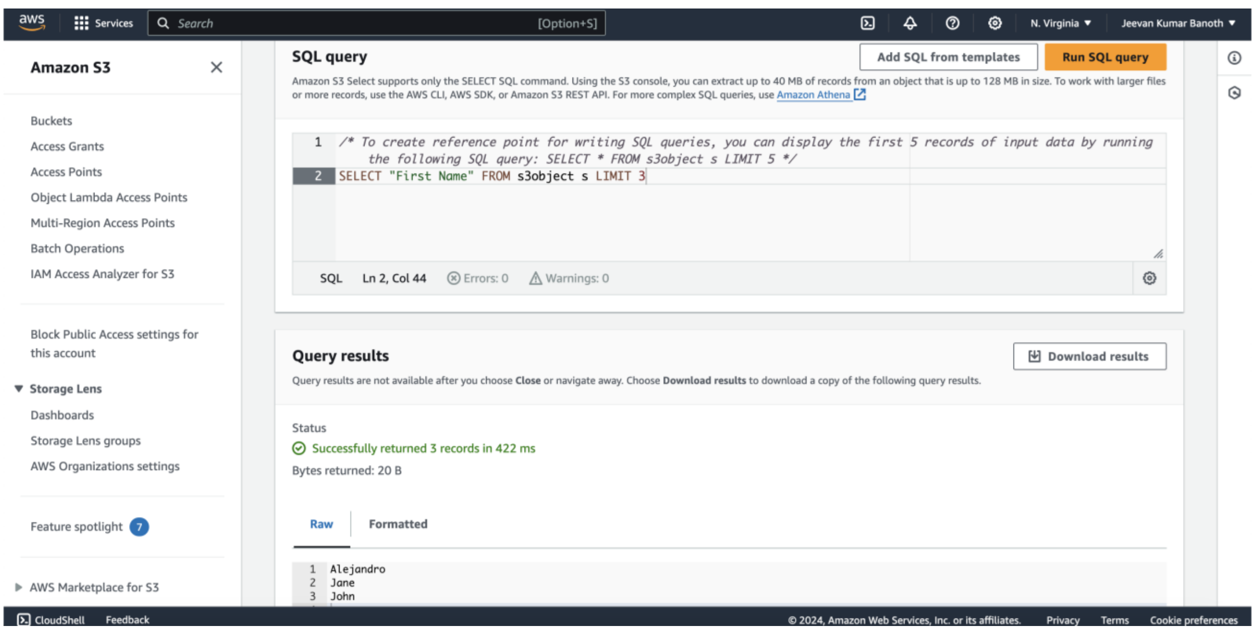
After this, in the input settings section we will be keeping the format as CSV selected. In the CSV delimiter keep the comma as selected and select the exclude the first line of the CSV data.

For compression keeping none selected.

For output configure setting keeping the default values as CSV and comma selected.



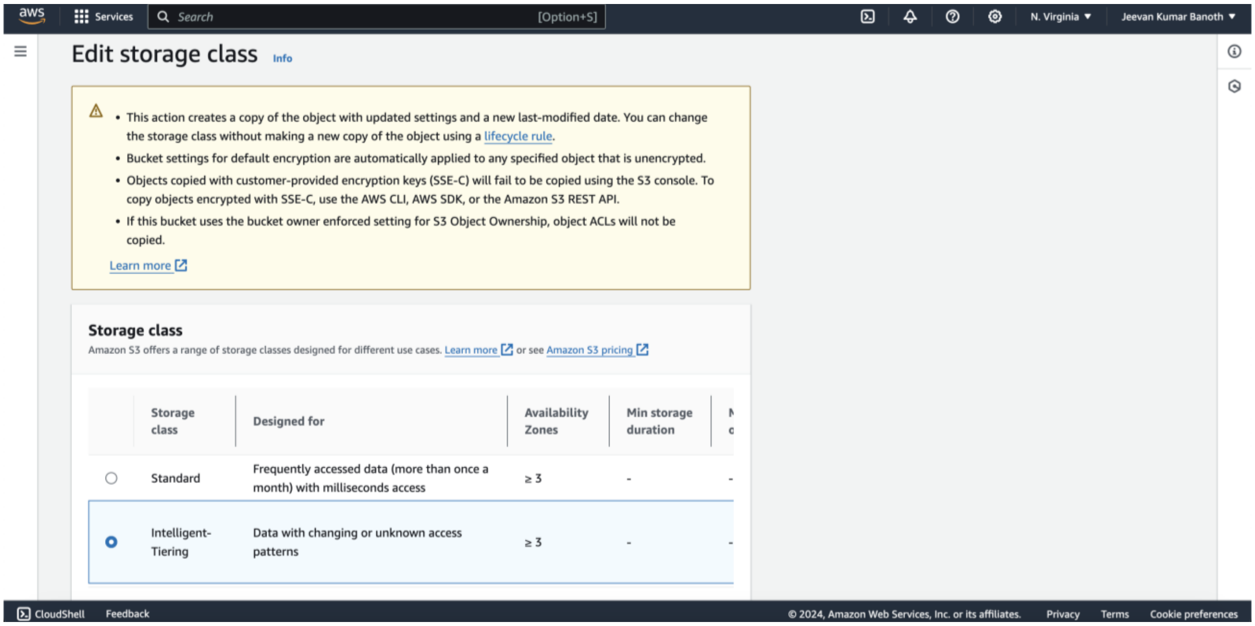
As shown above we need to run the SQL Query “SELECT * FROM s3object s LIMIT 5” as this, which will give all rows present in there.



First, we will be excluding the first line of CSV data that is selected in the Input Settings section. Then we will be changing the SQL Query in such a way that it will display only the first name of the people and we have limited it to only 3 rows.

Task 4: Modifying an Object's Encryption Properties and Storage Type: -

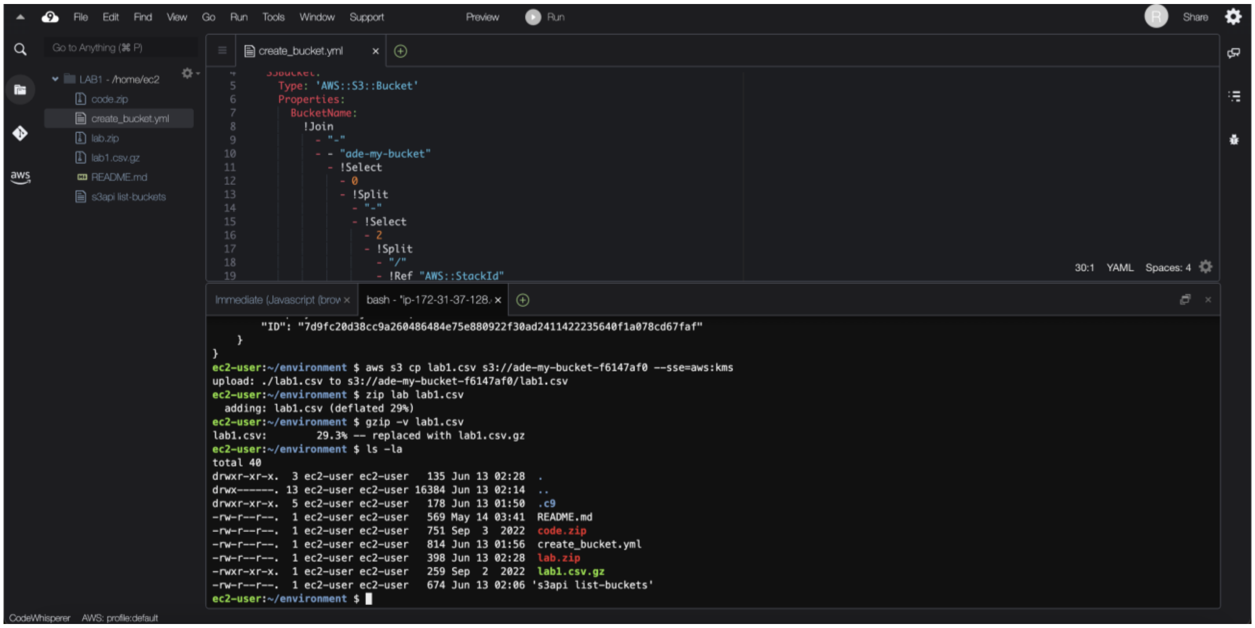
Now we will be modifying the object encryption properties by storage type in the edit section by choosing Intelligent-Tiering.



Task 5: Compressing and Querying the Dataset: -

Now, I will be compressing the data file by using AWS Cloud9 terminal and we will be zipping it. Then we will be g zipping the zipped data file.

To list the objects in the directory, we will be using “ls –la” and running it.



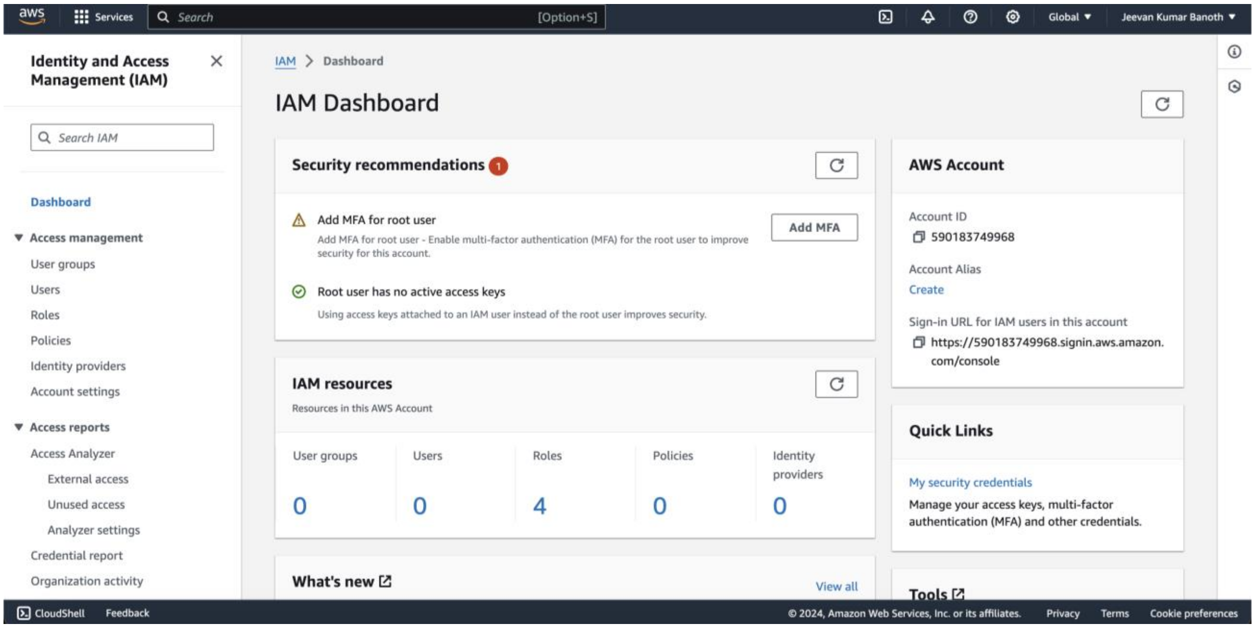
Now, we will be uploading the gzip file to the S3 bucket

Confirm Query with Compressed File

1. Return to the Amazon S3 console.
2. In the breadcrumbs at the top of the page, choose the bucket name.
3. Choose the lab1.csv.gz link.
4. Choose **Object actions** > **Query with S3 Select**.
5. In the **Input settings** section, for **Compression**, choose **GZIP**.

Task 6:

Managing and Testing Restricted Access for a Team Member:



For performing this above task which is managing and testing restricted access for Team members, I won't be able to do it because as I am performing this from my separate account, so I don't have the access to any of the groups.

This is the reason why I won't be able to perform the task from now on.

Summary: -

So, in this lab I have learned that:
how to create an S3 bucket using a cloud9 formation template to ensure that repeatable and automated process.

Added the data into S3 Bucket and performed few operations like creating stack, deleting and listing all stacks present in there.

Copying the data from the stack into bucket and viewing the contents of the dataset by using the cat commands.

Using the SQL Query to in the S3 select to view the contents in the dataset.

I got to know how to zip and unzip the dataset, which was downloaded, and how to perform the Gzip operations in that.

Ran the SQL Queries to run the specific queries.

I know how to modify the objects' encryption properties and storage type. Compressing and Quering the dataset.

So, from this lab I have learned all the above things.

----------X---------